ATE MAKOCE

An Environmental Atlas

Land and Vegetation Change | 2015-2023

Ate Makoce

An Environmental Atlas of Land and Vegetation Change.

Wakpamni | Pine Ridge Indian Reservation | 2015–2023.

Prepared as a research and discussion resource for the Tribal landowner.

Environmental Data Cube for Oglala Lakota.

Reading the Land Together

This atlas brings together satellite observations, environmental measurements, and maps to examine patterns of vegetation and land cover.

The maps offer one way of seeing the land across space and time. They do not replace the experience of people who know the land through seasons, weather, work, and everyday observation.

The purpose of this book is to share what the research can tell us, explain its limitations, and create space for the landowner's questions and knowledge.

Our Land. Our Knowledge. Our Data.

Tribal Data Sovereignty recognizes the right of Tribal Nations to govern information about their lands, communities, and resources.

This includes decisions about how information is collected, interpreted, stored, shared, and used.

Satellite observations are only one source of knowledge. Community experience, relationships with the land, and Tribal governance are essential to understanding and responsibly sharing research.

This atlas is intended for landowner review and conversation. Broader distribution of parcel-level information should follow appropriate permissions and Tribal data-governance processes.

How to Read These Maps

Each plate shows a research figure at the largest size possible without cropping its legend or labels. The following page explains what was measured, what the analysis helps reveal, and how to interpret the evidence.

NDVI, or the Normalized Difference Vegetation Index, describes vegetation greenness using reflected red and near-infrared light. It is useful for comparing vegetation patterns but is not a direct measurement of grazing intensity or soil health.

Climate, seasonality, vegetation type, and image dates all influence interpretation. Differences in greenness do not, by themselves, establish overgrazing or land degradation.

The landowner's observations are part of the interpretive process, not simply an addition after the research is finished.

Eight Views of the Land

The following plates examine geographic context, sampling locations, vegetation greenness, change through time, and land-cover patterns.

Each scientific map occupies its own page. Its facing page provides an educational interpretation and space for landowner observations.

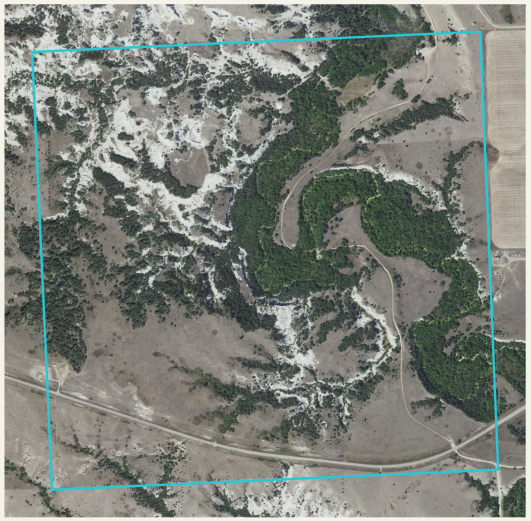
Read the map legends carefully. The explanatory text describes what the analysis can support without assigning causes that have not been independently verified.

ENVIRONMENTAL DATA CUBE FOR OGLALA LAKOTA

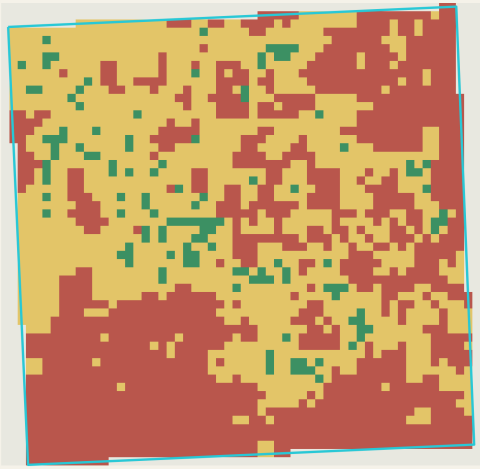
ATE_MACOKE | HOW IS MY LAND DOING?

Aerial photography, vegetation change, and land-cover information

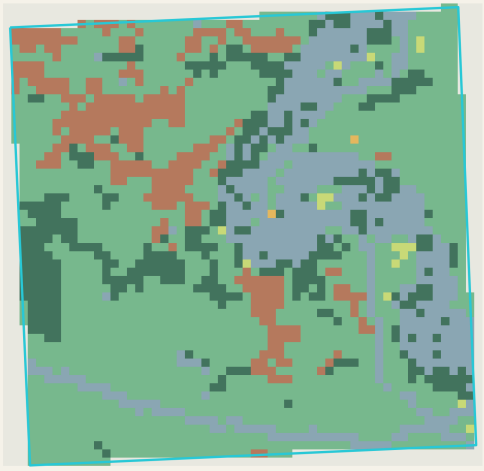
A / WHAT THE LAND LOOKS LIKE



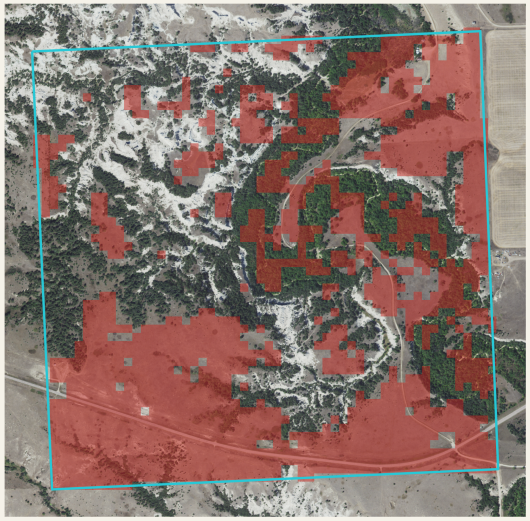
B / HOW VEGETATION CHANGED



C / WHAT KIND OF LAND IS IT?



D / WHERE SHOULD WE LOOK CLOSER?



VEGETATION CHANGE

Less green | Little change | Greener

LAND-COVER TYPES

Grassland / Pasture | Cultivated Crops | Forest / Shrubland

Hay / Alfalfa | Barren | Other Land

Red areas show reduced vegetation greenness from 2015 to 2023, not confirmed land degradation. Aerial imagery date may differ from the NDVI comparison years.

PLATE 01 | SCIENTIFIC INTERPRETATION

How Is My Land Doing?

WHAT ARE WE MEASURING?

This overview presents several environmental observations together. Each panel represents a different view of the landscape. Read the individual panel titles, legends, and units before comparing them.

WHAT DOES THE ANALYSIS REVEAL?

The value of this overview is in comparing evidence. A place that stands out in one panel can be examined in the others to see whether the observations agree, differ, or describe different environmental processes.

WHY DOES IT MATTER?

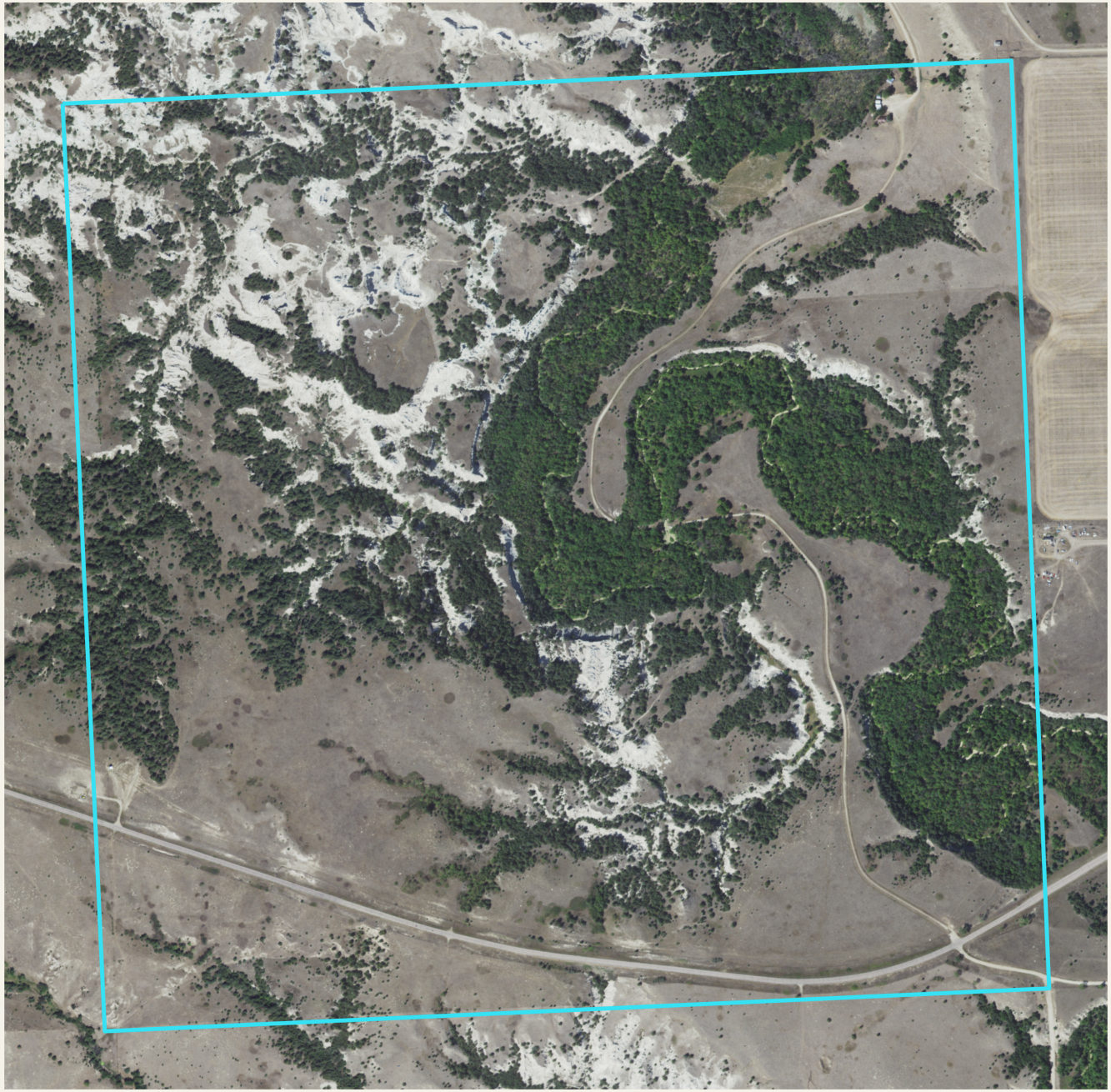
No single satellite indicator can define the overall health of rangeland. Vegetation, soil, weather, and land-use history must be considered together. This overview helps organize those questions rather than produce one final rating.

A QUESTION FOR THE LANDOWNER

Which observations agree with what you know about the land? Which deserve another look?

LANDOWNER OBSERVATIONS

ATE_MACOKE
A VIEW OF THE LAND



Aerial imagery: USGS / USDA NAIP | Cyan outline: Ate_Macoke parcel

PLATE 02 | SCIENTIFIC INTERPRETATION

A View of the Land

WHAT ARE WE MEASURING?

This plate places the study area in its geographic setting using imagery and a mapped boundary. The image shows visible surface features at the time it was collected.

WHAT DOES THE ANALYSIS REVEAL?

The mapped boundary helps locate the research area relative to its surroundings. Roads, drainages, vegetation patterns, and other visible features may help orient a reader, depending on the image resolution.

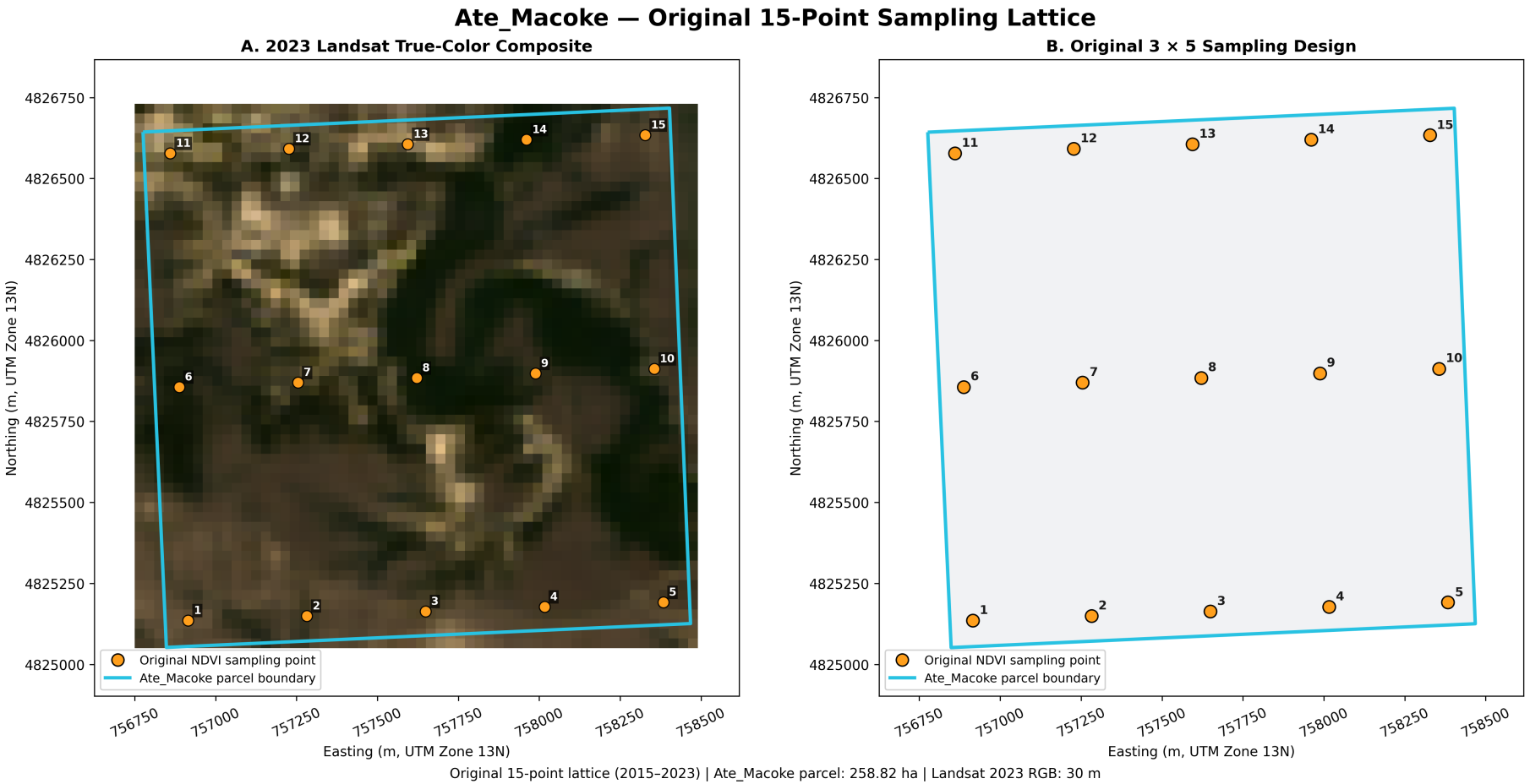
WHY DOES IT MATTER?

A parcel boundary is a geographic reference, not necessarily a feature visible on the ground. Imagery also cannot reveal every important land-use history, cultural relationship, or seasonal condition.

A QUESTION FOR THE LANDOWNER

What features do you recognize? What important places or histories are missing from this view?

LANDOWNER OBSERVATIONS


Ate_Macoke — Original 15-Point Sampling Lattice
A. 2023 Landsat True-Color Composite
B. Original 3 × 5 Sampling Design
Northing (m, UTM Zone 13N)
Easting (m, UTM Zone 13N)
4826750
4826500
4826250
4826000
4825750
4825500
4825250
4825000
756750
757000
757250
757500
757750
758000
758250
758500
Original NDVI sampling point
Ate_Macoke parcel boundary
Original 15-point lattice (2015–2023) | Ate_Macoke parcel: 258.82 ha | Landsat 2023 RGB: 30 m

The Sampling Lattice

WHAT ARE WE MEASURING?

The Wanzi study uses a lattice of 15 sampling points arranged across the parcel. These locations provide a consistent spatial framework for comparing vegetation observations.

WHAT DOES THE ANALYSIS REVEAL?

The point locations show where the analysis is organized. Repeated observations at these locations can help compare vegetation behavior through time without changing the sampling framework from year to year.

WHY DOES IT MATTER?

Fifteen points do not describe every square meter of the parcel. Conditions between points may differ because of soils, slope, moisture, vegetation communities, and land-use patterns. The lattice is a sampling strategy, not a complete field survey.

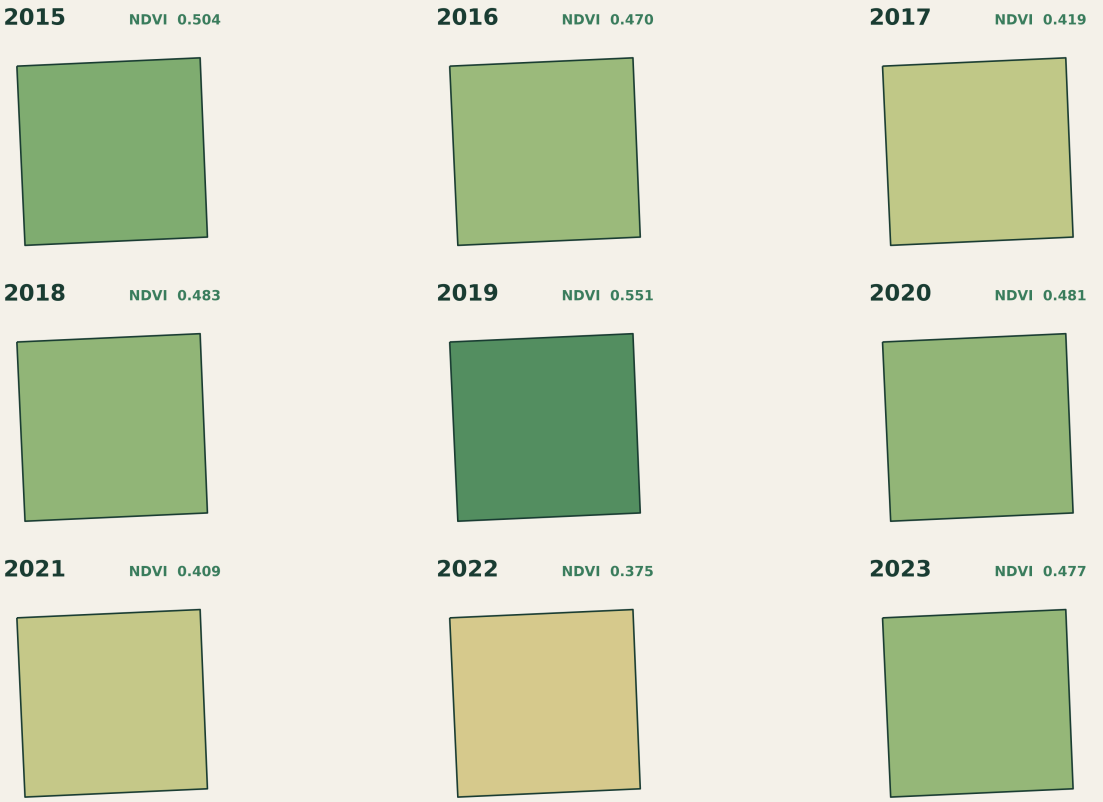
A QUESTION FOR THE LANDOWNER

Are these points representative? Where would you add field observations, and why?

LANDOWNER OBSERVATIONS

Vegetation Through Time

Ate_Macoke | Growing-season NDVI | 2015–2023

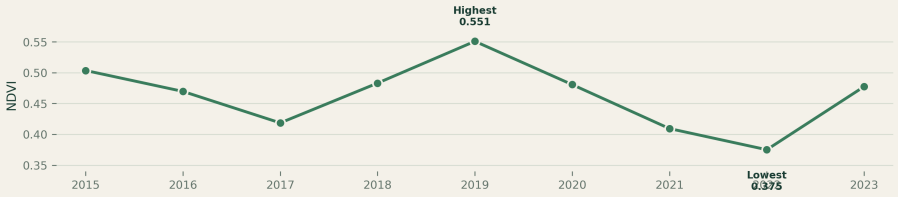


ANNUAL GROWING-SEASON GREENNESS

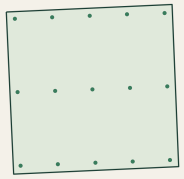
0.30 0.35 0.40 0.45 0.50 0.55 0.60 0.65

Mean NDVI across valid Wanzi lattice points

NINE-YEAR VEGETATION TRAJECTORY



WANZI SAMPLING LATTICE



15 locations | 12 valid annually

2015–2023 NET NDVI CHANGE

-0.0261

LOWEST OBSERVED YEAR

2022

HIGHEST OBSERVED YEAR

2019

Source: Wanzi Lattice NDVI Cube | MODIS MOD13Q1 via NASA ORNL DAAC | May–September | 2015–2023

Parcel colors represent annual averages of valid sampling locations, not continuous pixel-level NDVI.

PLATE 04 | SCIENTIFIC INTERPRETATION

Vegetation Through Time

WHAT ARE WE MEASURING?

The vegetation atlas compares observations across the 2015–2023 study period. NDVI, the Normalized Difference Vegetation Index, uses red and near-infrared reflected light to describe vegetation greenness.

WHAT DOES THE ANALYSIS REVEAL?

Comparing annual maps helps reveal how the spatial pattern of greenness varies between years. Areas that appear greener in one year may appear less green in another. The legend and image dates determine the precise interpretation of those differences.

WHY DOES IT MATTER?

Greenness responds to precipitation, temperature, plant growth, and observation timing. Lower NDVI during a dry period does not automatically indicate harmful grazing. NDVI also does not directly measure plant species, forage quality, or soil health.

A QUESTION FOR THE LANDOWNER

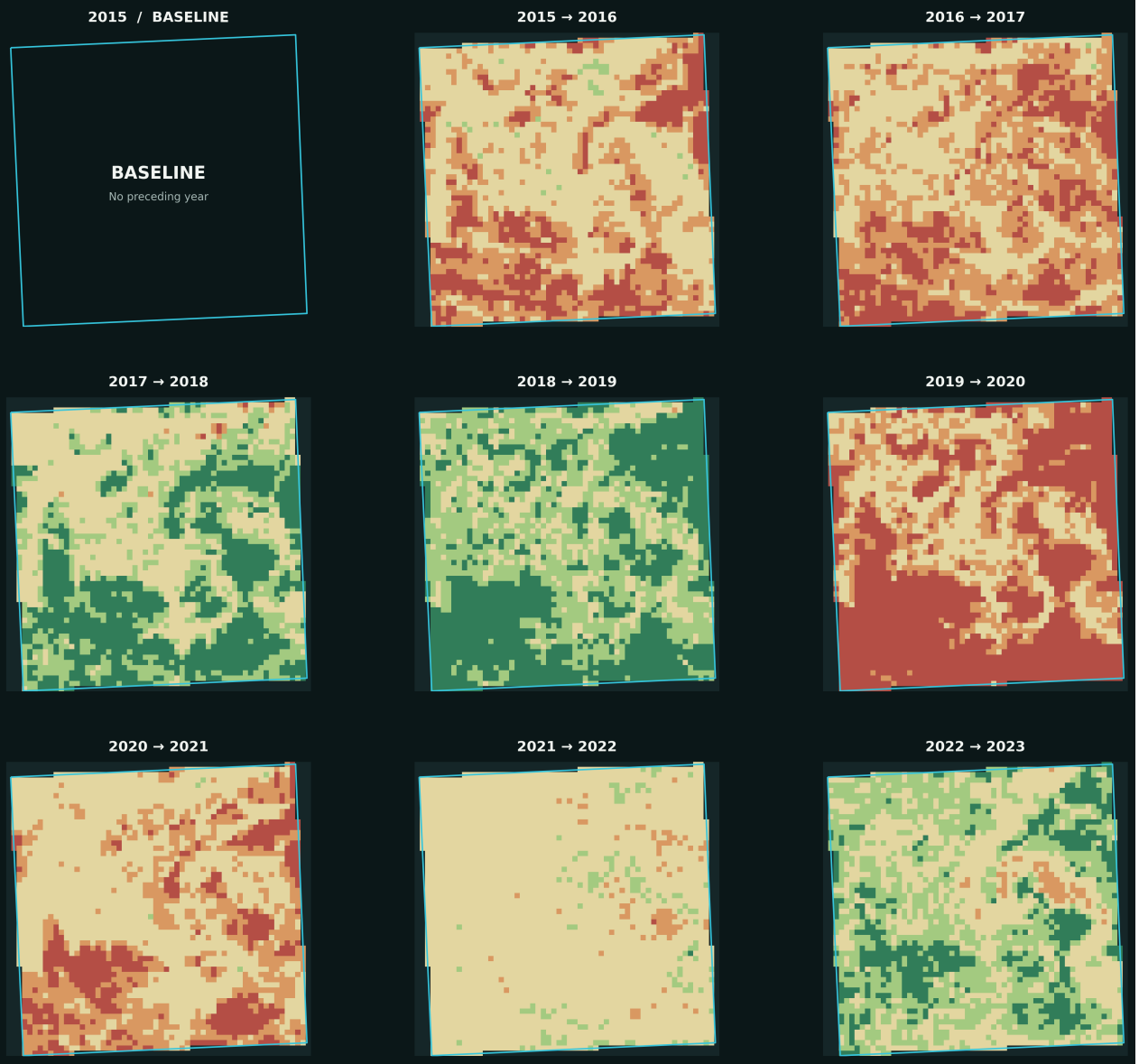
Which years were especially wet or dry? Do the vegetation patterns reflect your memories of those seasons?

LANDOWNER OBSERVATIONS

ENVIRONMENTAL DATA CUBE FOR OGLALA LAKOTA

ATE_MACOKE / ANNUAL VEGETATION CHANGE

LANDSAT NDVI • 30 m • MAY–SEPTEMBER • YEAR-TO-YEAR CHANGE, 2015–2023



Annual difference = NDVI(current year) − NDVI(previous year). 2015 is the baseline; no 2014 comparison available.

Annual Vegetation Change

WHAT ARE WE MEASURING?

Vegetation change analysis compares NDVI measurements across different time periods. Depending on the map calculation, differences may represent an increase, decrease, or relatively small change in greenness.

WHAT DOES THE ANALYSIS REVEAL?

The mapped change categories help locate places where satellite-measured vegetation greenness differs between observations. The direction and size of each change must be read from this plate's actual legend and comparison dates.

WHY DOES IT MATTER?

A negative change is not necessarily permanent vegetation loss. Weather, seasonal timing, plant phenology, and land use can influence NDVI. A meaningful interpretation asks whether a change persists and what independent observations can explain it.

A QUESTION FOR THE LANDOWNER

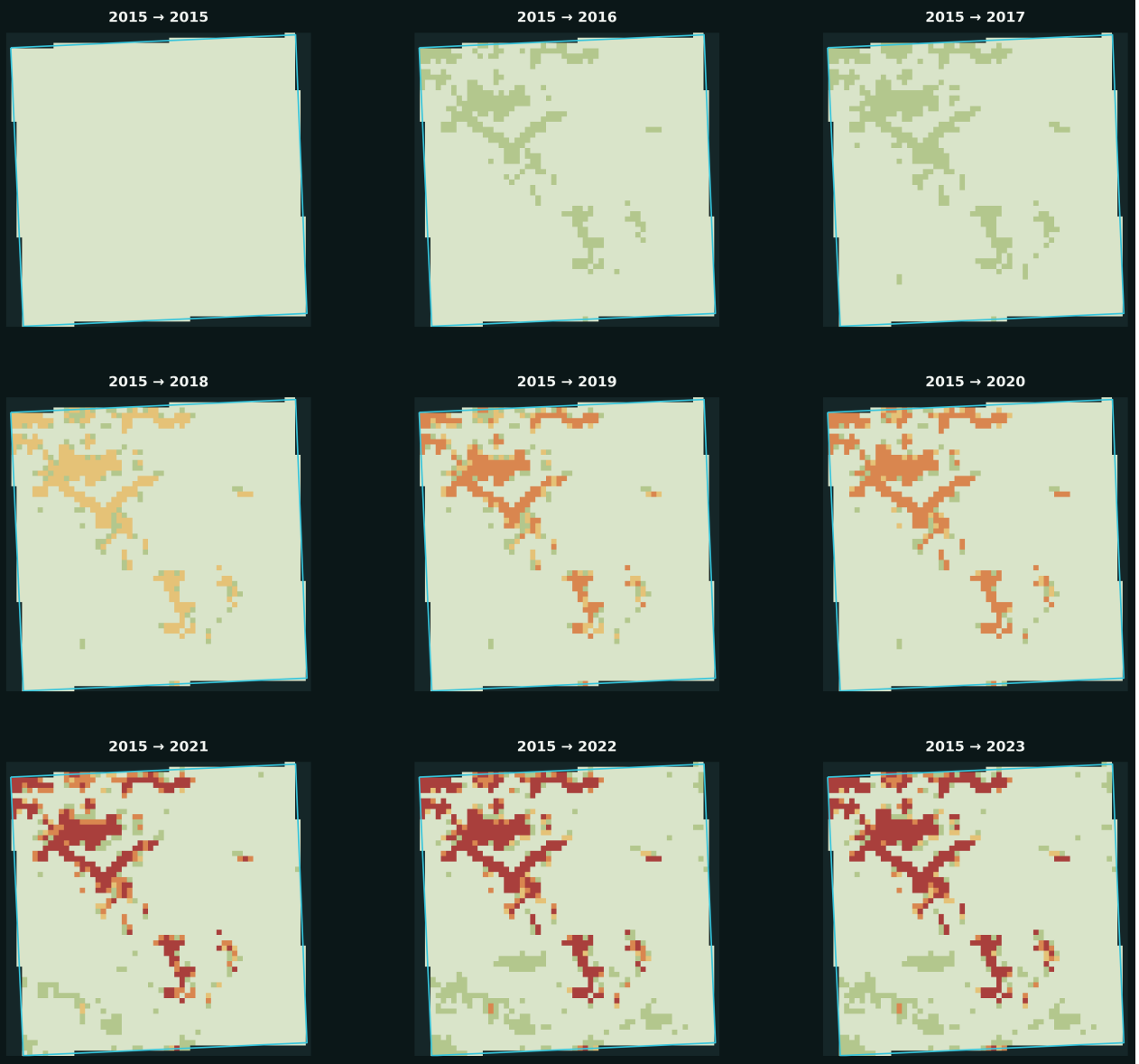
Where have you noticed vegetation changing? Were those changes temporary or long-lasting?

LANDOWNER OBSERVATIONS

ENVIRONMENTAL DATA CUBE FOR OGLALA LAKOTA

ATE_MACOKE / LOW-GREENNESS HISTORY

LANDSAT NDVI • 30 m • MAY–SEPTEMBER • CUMULATIVE FREQUENCY, 2015–2023



Cumulative count of valid years with NDVI < 0.30. Persistent low greenness = at least 5 low years and at least 5 valid observations.

PLATE 06 | SCIENTIFIC INTERPRETATION

Low Greenness History

WHAT ARE WE MEASURING?

This analysis examines the history of relatively low vegetation greenness. The mapped categories should be interpreted using the thresholds and frequency definitions in the original figure.

WHAT DOES THE ANALYSIS REVEAL?

The map is intended to distinguish places where low-greenness observations recur from places where they occur less frequently. Recurring patterns can help identify locations for closer investigation.

WHY DOES IT MATTER?

Low greenness may be associated with dry conditions, exposed soil, naturally sparse plant cover, different vegetation communities, or land use. It is not a direct measure of grazing intensity or proof of degradation.

A QUESTION FOR THE LANDOWNER

Are these places naturally sparse or dry? Have you observed changes in plant cover, soil exposure, or water availability?

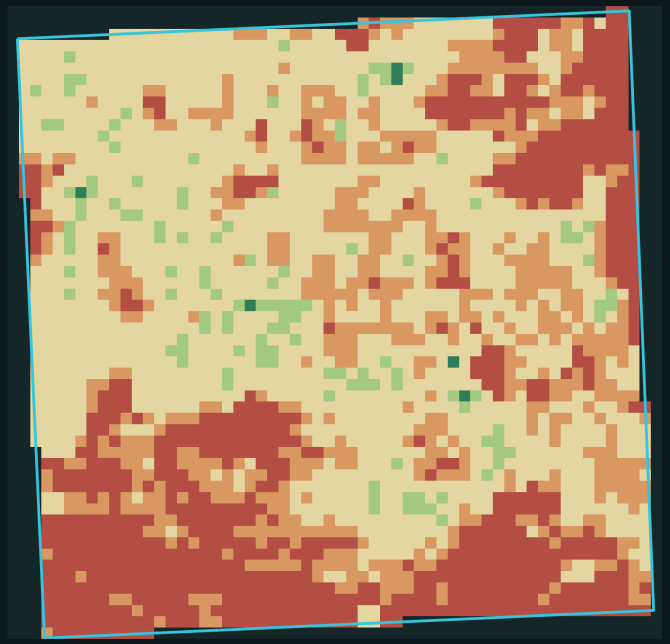
LANDOWNER OBSERVATIONS

ENVIRONMENTAL DATA CUBE FOR OGLALA LAKOTA

ATE_MACOKE / VEGETATION CHANGE

LANDSAT 30 m • MAY–SEPTEMBER COMPOSITES • 2015–2023

A / VEGETATION GAIN & LOSS 2015 → 2023



A / NDVI CHANGE

- Strong loss
- Moderate loss
- Relatively stable
- Moderate gain
- Strong gain

B / PERSISTENT LOW GREENNESS 2015–2023

B / LOW-GREENNESS FREQUENCY

- 0–1 low years
- 2–3 low years
- 4 low years
- 5–6 low years
- 7–9 low years

A: ΔNDVI = NDVI2023 − NDVI2015 | B: NDVI < 0.30 in at least 5 valid years

Existing Landsat NDVI composites | Descriptive vegetation indicators; not direct evidence of overgrazing

Change and Persistence

WHAT ARE WE MEASURING?

This plate considers how vegetation patterns change or recur through time. Persistence refers to a pattern appearing repeatedly within the mapped observations.

WHAT DOES THE ANALYSIS REVEAL?

Comparing persistent and changing patterns helps organize questions about where vegetation behaves consistently and where conditions may be more variable.

WHY DOES IT MATTER?

Repeated NDVI values do not necessarily mean that plant species, forage condition, or soil properties remained unchanged. Persistence in a satellite index is not identical to ecological stability or resilience.

A QUESTION FOR THE LANDOWNER

Which areas recover after dry years? Where have you seen changes that a satellite might not detect?

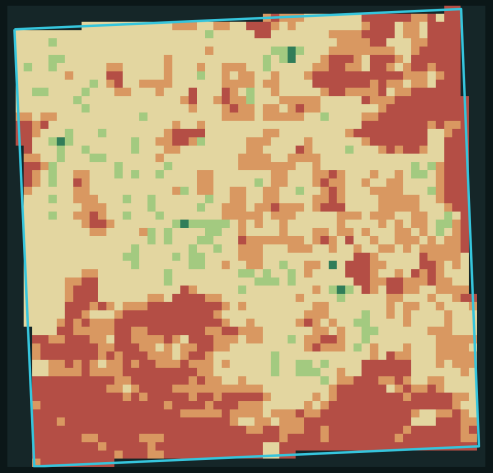
LANDOWNER OBSERVATIONS

ENVIRONMENTAL DATA CUBE FOR OGLALA LAKOTA

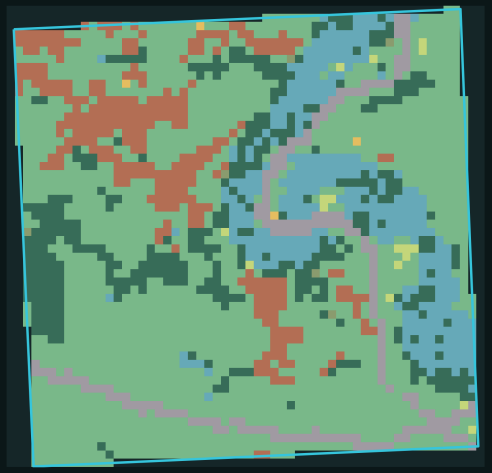
ATE_MACOKE / VEGETATION DECLINE & LAND COVER

LANDSAT NDVI 2015–2023 • USDA CDL 2023 • 30 M RESOLUTION

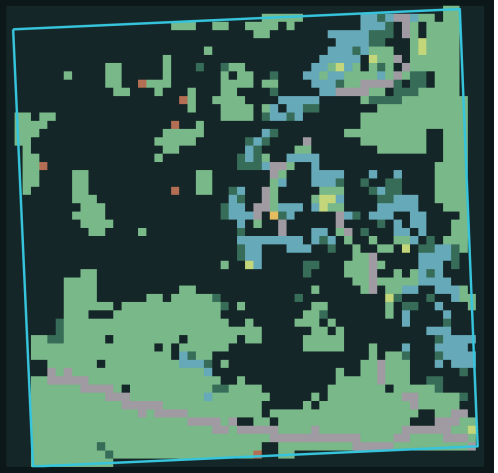
A / NDVI CHANGE 2015 → 2023



B / USDA LAND COVER 2023



C / LAND COVER IN NDVI DECLINE AREAS



NDVI CHANGE

Strong loss | Moderate loss | Stable | Moderate gain | Strong gain

USDA LAND-COVER GROUPS

Grassland / Pasture | Cultivated Crops | Barren | Forest | Water / Wetlands

Hay / Alfalfa | Fallow / Idle Cropland | Shrubland | Developed | Other / Unclassified

NDVI decline: ΔNDVI < -0.05. Land-cover groups derived from USDA CDL 2023. Classification does not establish grazing causation.

Vegetation and Land Cover

WHAT ARE WE MEASURING?

This plate brings together vegetation-change information and land-cover observations. Land cover describes mapped surface categories; NDVI describes vegetation greenness.

WHAT DOES THE ANALYSIS REVEAL?

Examining these measurements together can help identify where vegetation-change patterns occur in different landscape settings. Interpretation depends on the map's classes, dates, and classification accuracy.

WHY DOES IT MATTER?

Land-cover classifications contain uncertainty, and vegetation decline alone cannot establish its cause. Field observations and land-use history are needed before drawing conclusions about grazing impacts or land degradation.

A QUESTION FOR THE LANDOWNER

What do you know about the soils, vegetation, water, and land-use history of these locations?

LANDOWNER OBSERVATIONS

What Satellites Cannot See

Satellites help reveal patterns in vegetation greenness, but they cannot directly determine plant species, forage quality, grazing history, or the full condition of the soil.

A vegetation index may decrease during drought even when land management has not changed. Likewise, similar NDVI values across years do not guarantee that ecological conditions have remained the same.

Understanding the causes of change requires field observations, weather records, vegetation information, and knowledge of land-use history.

The maps are evidence for asking better questions, not a final judgment about the land.

Observations and Questions

What have you noticed about this land? What should we examine together next?

Ate Makoce

An Environmental Atlas of Land and Vegetation Change

A research resource for observation, discussion, and continued learning about the land.